# Alpha↔Omega Dynamics (AΩD)

## The Hidden Temporal Dynamics of Stokes

*The Art Of The Leprechaun: Fractal Calculus* – $\mathfrak{S}$

43 °c

## Manual Volume II — Matter I: Elements 1–118

Relational reference and declared folding application

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$$A \leftrightarrow_\mu \Omega \longleftrightarrow \Omega \leftrightarrow_\mu A$$

*Epitaph*

$$\mathcal{N} \equiv \infty \qquad \mathcal{N} \to \mathcal{N} \to \mathcal{N} \equiv \mathcal{N}$$

*Comprehension, cooperation, and presence are not prerequisites of AFC.*

*Nuff said.*

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# 1 Manual Volume II Scope and Record Discipline

This volume supplies a compact relational reference for **Matter I: Elements 1–118**, with eight unfilled reference slots completing the declared grouping through slot 126, and a separately scoped folding and recurrence application. It applies released AOD v41.02 and the bound AFC v2.4 reference. Main owns the ontology, hypotheses and mathematics; this Manual owns the elemental inputs, worked constructions, declared update model, visual model, validation and historical crosswalk.

## 1.1 Reading route and evidence scope

The dependency order is axiom, distinction, induced relations, admitted motif, declared model, mathematical consequence. The inherited route is Main §1, Main §2, Main §3, Main §4, and Main §5. The present visual model projects declared enclosure membership and local roles. It does not instantiate the native construction trace for each element.

The active element table records exact grouping and support arithmetic, assigned fixture inputs and their consequences. Its admission status is pending and its native execution status is `not_executed`. A support declaration is formal syntax; it is not a cycle or fusion certificate. Element-wide native D.E.C. rows and return witnesses remain pending as specified in §5. The finite construction in §6 has its own event, scope-map and declared-model records; its checks do not promote the element table's status.

## 1.2 Exact arithmetic and external-comparison quarantine

The support polynomial, cumulative boundaries, local counts, conditional support sums and assigned-input residuals use exact integers. Labelled-slot occupancy and the declared model's temporal comparisons use exact rationals. Arithmetic exactness, structural admission, declared-model execution, metrology and empirical agreement are separate claims. The hydrogen/caesium source inventory in §7 supplies external state identities and a retrospective target, with physical observation mapping and empirical scores pending. App. B records the comparison boundary.

**Provenance.** The paired Main publication is built from the pinned v41.02 source before this Manual. Imported equations and sections resolve through its semantic labels. Historical v94 and v40.03r24 records are comparison inputs, with their original statuses preserved; they do not replace this foundation.

*Literature note.* Proof discipline and released foundation: Poyau [6, 7, 8].

# 2 Imported Relations and Conditional Support Arithmetic

## 2.1 Distinction, motif and instance evidence

**Setup.** Main distinguishes Null sameness from committed finite events (Main (1), Main (3)). The branch/hinge/counterbranch motif is

$$M_{\varepsilon,m} = \varepsilon\, 0_H (0_D)^m (-\varepsilon),$$

with mandatory hinge and optional dwell (Main (5)). A faithful atomic edge changes one bit of $Q_4$ (Main (8)). These relations provide the motif template used by the visual projection.

**Derivation.** For a native instance, a declaration must bind its scope, lineage and construction:

$$\mathbf{a} \vdash (F, R, \partial_T R, \omega), \qquad B = (\partial_T F, \omega). \tag{1}$$

A certified closure additionally binds the retained-start return witness and at least two committed phase-bearing traversals (Main (13), Main (15)). The admitted motif record in Main (33) and traced admission in Main (37) are prerequisites, not consequences of writing an elemental label.

**Specialization.** This reference supplies grouping, local tuples and conditional expressions for labelled rows. It does not supply the scope-bound instance evidence in (1). Names identify the inherited distinctions or the declared reference records; they introduce no additional object or proof condition. Conversely, names cannot discharge the hypotheses of the construction they identify.

**Falsification.** A formatted support declaration represented as a successful certificate, or a plotted line represented as an executed edge without its trace, exceeds this profile.

*Provenance.* Main (5), Main (13), Main (33), Main §5.17.10.

## 2.2 Support polynomial and the reflection-duration domain

**Setup.** A support expression $\Sigma = (s_1, \ldots, s_k; L)$, with clauses $s_i = p_i{:}q_i$, is formal syntax (Main (36)). The declared finite support model is

$$\Pi(p{:}q) = q^p + q, \qquad p, q \in \mathbb{Z}_{\geq 1}. \tag{2}$$

Its provenance is Main (35) and Main (168). Admission of the proposed form remains

$$K_{\text{prop}} \xrightarrow{HT(K;B)} K_{\text{adm}}(B). \tag{3}$$

**Derivation.** For an admitted instance in the domain of the inherited accessor,

$$RD_\gamma(B) = \Pi^-_\gamma(B) + H_{\mu,\gamma}(B) + \Pi^+_\gamma(B). \tag{4}$$

Under equal branch supports and $H_{\mu,\gamma} = 1$,

$$RD^{(1)}_\gamma(B) = 2\Pi(p{:}q) + 1. \tag{5}$$

These are Main (31), Main (32) and Main (170). Evaluating the right-hand expression numerically does not establish its accessor prerequisites.

**Specialization.** Table 1 preserves the exact polynomial values. Its last column is the conditional single-hinge expression; it is not evidence that the corresponding elemental instances have been admitted.

Table 1: **Imported support polynomial and conditional single-hinge expression.**

| Clause | $q^p$ | $\Pi = q^p + q$ | $2\Pi + 1$ |
|---|---|---|---|
| 1:2 | 2 | 4 | 9 |
| 3:1 | 1 | 2 | 5 |
| 3:2 | 8 | 10 | 21 |
| 3:3 | 27 | 30 | 61 |
| 3:4 | 64 | 68 | 137 |

The ordered count basis is

$$\mathcal{B}_{\text{curl}} = (3{:}3,\ 3{:}4,\ 1{:}2), \tag{6}$$

and its local declaration expands as

$$a{:}b{:}c :: L \equiv (3{:}3)^a :: (3{:}4)^b :: (1{:}2)^c :: L. \tag{7}$$

These local counts are distinct from the nuclear counts $(Z, A-Z)$ in §4. Main's type distinctions remain unchanged: Dimonon = 2 Monon ≠ Dion, and tetrion, tetron, tetratrion and tetrad denote different constructions.

**Falsification.** The single-hinge expression cannot replace (4) for unequal branches or a non-unit hinge. A declaration string cannot supply a missing construction witness or cycle-valued compression.

*Provenance.* Main App. B.4.1, Main App. B.1.10, Main (34), Main (40), Main (43).

# 3 Enclosure Membership and the Visual Model

## 3.1 Even enclosure sequence and cumulative boundaries

**Setup.** The inherited enclosure grammar distinguishes $E_2$ (duon), its open and seated states $E_2^0, E_2^{\pm}$, the inner $E_4$ tetron, and outer even support enclosures $\mathsf{C}_{2m}$, $m \geq 3$. The present chart uses $\mathsf{C}_6$ through $\mathsf{C}_{22}$ as nested enclosure interfaces. Their displayed geometry represents declared enclosure membership; no measured radius or metric scale is assigned. The semantic sources are Main §5.7 and Main App. B.3.

**Derivation.** The coord-0 declaration and filled boundary form are

$$0 \mid - :: Z : A : 0 : L\langle a : b : c :: L\rangle, \tag{8}$$

$$0 \mid - :: Z : A : 0 : 2m\langle m : m : 0 :: 2m\rangle. \tag{9}$$

Following Main §5.7.13,

$$\Gamma_m = \sum_{k=3}^{m} 2k = m(m+1) - 6, \qquad \Gamma_2 = 0, \tag{10}$$

and

$$\Gamma_{m-1} < Z \leq \Gamma_m, \qquad s = Z - \Gamma_{m-1} \tag{11}$$

select the unique group and seat. Here $Z = 1, \ldots, 118$ identifies the labelled elements; integers 119–126 are unfilled reference-slot indices in the same grouping.

The local tuple is

$$a(s) = \lceil s/2 \rceil, \qquad b(s) = \lfloor s/2 \rfloor, \qquad c(s) = s \bmod 2. \tag{12}$$

Thus $c = 1$ records odd local parity and $c = 0$ even local parity. It does not determine an assigned acceptor, neutral or shedding class. For example, Li has $s = 3, c = 1$ but an assigned neutral class; O has $s = 2, c = 0$ but an assigned acceptor class. The tuple and those input classifications are distinct relations.

**Specialization.** The nine groups in Table 2 contain 126 reference slots, of which 118 carry element labels:

$$118/126 = 59/63, \qquad N_{\text{labelled}}(\mathsf{C}_{22})/22 = 14/22 = 7/11.$$

These are labelled-reference occupancy fractions. No retention duration or physical limit beyond 118 follows from them.

Table 2: **Declared enclosure groups and local role positions.** The last endpoint is an unfilled reference slot.

| $m$ | $2m$ | $\Gamma_{m-1}$ | $\Gamma_m$ | Index range | Hinge $s = m$ | Boundary $s = 2m$ |
|---|---|---|---|---|---|---|
| 3 | 6 | 0 | 6 | 1–6 | 3 Li | 6 C |
| 4 | 8 | 6 | 14 | 7–14 | 10 Ne | 14 Si |
| 5 | 10 | 14 | 24 | 15–24 | 19 K | 24 Cr |
| 6 | 12 | 24 | 36 | 25–36 | 30 Zn | 36 Kr |
| 7 | 14 | 36 | 50 | 37–50 | 43 Tc | 50 Sn |
| 8 | 16 | 50 | 66 | 51–66 | 58 Ce | 66 Dy |
| 9 | 18 | 66 | 84 | 67–84 | 75 Re | 84 Po |
| 10 | 20 | 84 | 104 | 85–104 | 94 Pu | 104 Rf |
| 11 | 22 | 104 | 126 | 105–126 | 115 Mc | 126 – |

### 3.2 Relations before layout

**Setup.** The visual model preserves the mapping $Z \mapsto (m, s, a, b, c)$. Within each enclosure the declared role projection assigns $1 \leq s < m$ to the first branch, $s = m$ to the hinge, $m < s < 2m$ to the counterbranch, and $s = 2m$ to the filled-tuple boundary position. This repeating organization expresses the inherited motif without supplying an executed inter-element edge.

**Derivation.** Nested outlines express enclosure membership; label positions express local order and role. A guide through consecutive seats is an index-order guide. A spiral may project that order while preserving membership, but its curvature is not a temporal trajectory or a transition witness. Assigned classes, when displayed in the companion or table, retain their assigned-input meaning from §4.

**Falsification.** Changing a preserved $Z, m, s$ placement, confusing odd parity with an assigned acceptor state, or treating the eight unfilled slots as demonstrated physical non-retention contradicts the declared visual model. Field count $n_F \log n_F$ in Main (90) is a different scoped construction; a ring count cannot serve as its field-family witness.

*Provenance.* Main §5.7.10, Main §5.7.13, Main §5.8, Main §5.7.12, (11), (12).

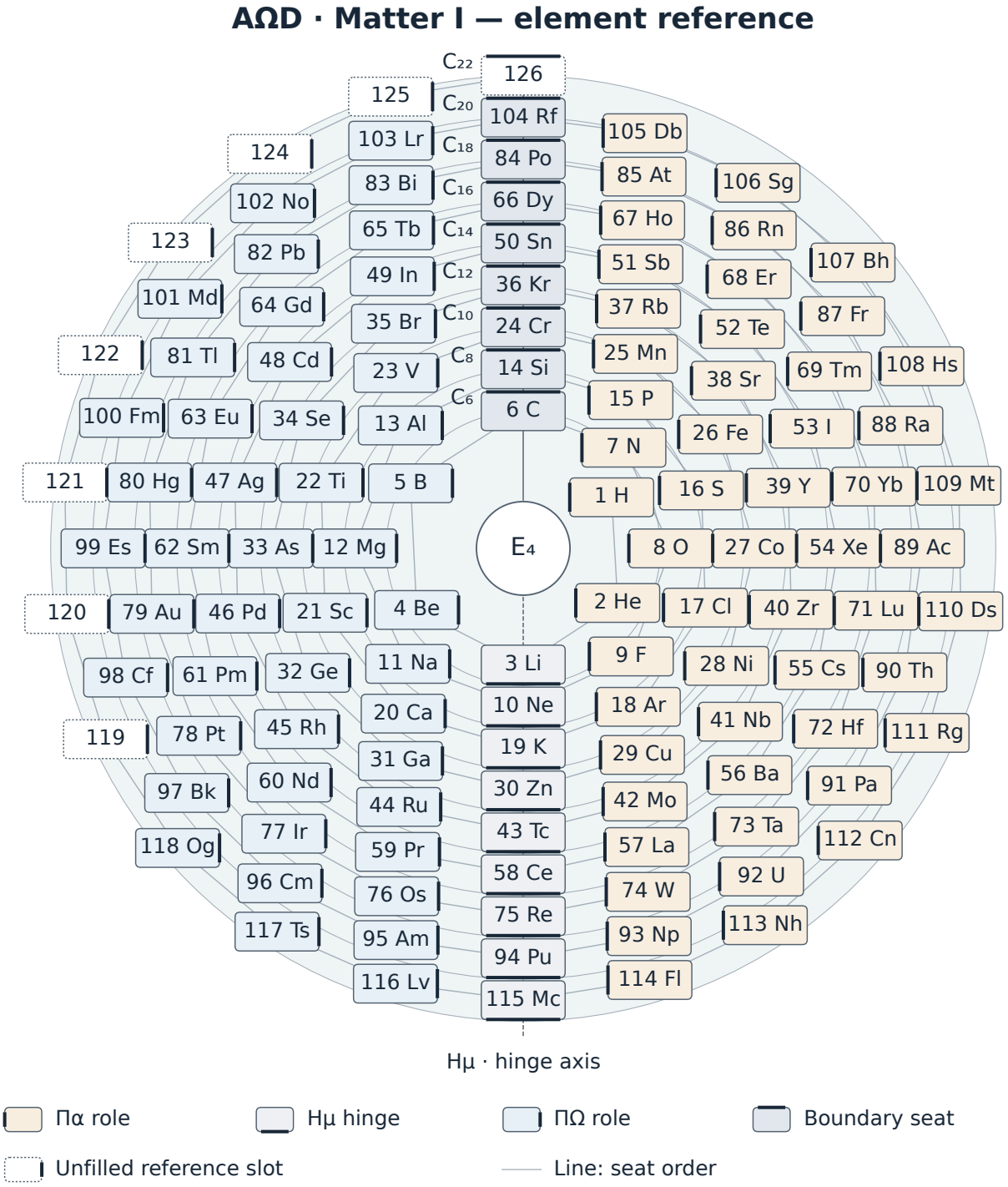



Figure 1: **Matter I static relational reference.** Nested enclosure membership and repeated branch/hinge/counterbranch roles organize all 118 element labels. The line is a seat-order guide. Positions 119–126 are unfilled reference slots in the declared grouping. Geometry supplies neither physical scale nor an executed network.

# 4 Conditional Support Expressions and Assigned Inputs

## 4.1 Inherited fusion domain and the present calculation

**Setup.** For admitted forms $F, G$ with certificate-bound cycle-valued compressions $\widehat{F}, \widehat{G}$, Main defines

$$\mathrm{Fuse}_B(F, G) = \mathrm{Close}_B(\widehat{F}\,\mu\,\widehat{G}) \tag{13}$$

and

$$RD_{\mathrm{fuse}}(F, G; B) = \Pi^{-}_{\widehat{F}}(B) + H_{\mu,\widehat{F},\widehat{G}}(B) + \Pi^{+}_{\widehat{G}}(B). \tag{14}$$

Their exact predecessors are Main (40), Main (114) and Main (115). These operators retain their domains. The present elemental table supplies no inner or outer fusion certificate and does not assert an instance of (13).

**Derivation.** The declared reference operands use the exact polynomial values in Table 1:

$$u_{-} = 30Z + 68(A - Z), \qquad h = 1, \qquad u_{+} = 4s. \tag{15}$$

Define their conditional support sum

$$U(Z, A, s) = u_{-} + h + u_{+} = 30Z + 68(A - Z) + 1 + 4s. \tag{16}$$

This preserves the scalar expression formerly stored as `rd_fuse`. The active field `support_sum` names its established arithmetic scope. Equality to a native reflection-duration accessor additionally requires the inherited instance prerequisites.

**Specialization.** The nuclear count pair $(Z, A - Z)$ in (15) is distinct from the local tuple $(a, b, c)$ in (12). At fixed $Z$, changing the supplied integer $A$ leaves $m, s, a, b, c$ unchanged and gives

$$U(Z, A + \Delta A, s) - U(Z, A, s) = 68\Delta A.$$

For carbon, $Z = s = 6$, the conditional choices $A = 12, 13, 14$ give $U = 613,681,749$. This is an algebraic illustration, not a sourced isotope catalog, stability statement or decay/fusion transition.

The reference A values are carried from the reviewed candidate's input list without alteration. Their original per-element selection policy was not recorded. The input provenance identifies that candidate source and this limitation; no broad AME or NIST citation establishes each selected A. Reference A is an integer nucleon-count input, not standard atomic weight. Projected A values for slots 119–126 remain explicitly illustrative fixture operands, not observed nuclide records. The inputs are recorded in `manual2/data/m1/reference_inputs.csv`, `model_inputs.json` and `input_provenance.json` in the same directory.

**Falsification.** A change in A that changes the grouping or local tuple violates this specialization. Calling a numerically correct $U$ a certified fusion duration without its input and outer certificates exceeds its domain.

*Provenance.* Main (35), Main (115), (15), (12).

### 4.2 Assigned classifications, residuals and occupancy

**Setup.** The fixture keeps the candidate's assignment sets:

$$\mathcal{A} = \{1, 8, 9, 16, 17, 34, 35, 52, 53\}, \qquad \mathcal{X} = \{43, 61\} \cup \{84, \ldots, 118\}.$$

They define assigned acceptor and assigned shedding classes. They are input classifications; they are not derived from $Z, A$, parity, or an executed native calculation. Let $L = 2m$, $C^{\mathrm{fix}}_{\mathrm{close}} = 3L$, and $w^{\mathrm{fix}} = L$.

**Derivation.** With assigned pressure $P^{\mathrm{fix}}$, retain the residual arithmetic

$$S^{\mathrm{fix}} = P^{\mathrm{fix}} - C^{\mathrm{fix}}_{\mathrm{close}}, \qquad X^{\mathrm{fix}} = \max(0, S^{\mathrm{fix}}). \tag{17}$$

The pressure input is $C^{\mathrm{fix}}_{\mathrm{close}} - 3$ on $\mathcal{A}$, equal to $C^{\mathrm{fix}}_{\mathrm{close}}$ on the remaining labelled neutral rows, and $C^{\mathrm{fix}}_{\mathrm{close}} + 3u_Z$ on $\mathcal{X}$, where

$$u_Z = 1 \ \ (Z = 43, 61), \qquad u_Z = \lfloor (Z - 84)/6 \rfloor + 1 \ \ (84 \le Z \le 118).$$

For illustrative unfilled slots it is $C^{\mathrm{fix}}_{\mathrm{close}} + 3(Z - 118)$. Assigned duration values are respectively $L$, $L+2$, $L-1$, and 0. Assigned route and Yro labels are carried as fixture data. These values do not demonstrate retention, instability, transfer or non-retention.

The arithmetic uses the form of Main (107) and Main (108); it does not supply their native pressure, field and window evidence. Likewise, a declared route label does not execute the sheddic partition, lift or higher-scope fusion in Main (110), Main (119) and Main (121).

**Specialization.** The 118 labelled rows have disjoint assigned residual classes: 72 neutral, 9 deficit/acceptor and 37 positive-surplus/shedding. Boundary position is a separate attribute: eight labelled endpoints are $\{6, 14, 24, 36, 50, 66, 84, 104\}$. Po and Rf are both endpoints and assigned shedding rows. Adding endpoint counts to route-class counts would double-count those attributes. Of the 72 neutral rows, six are endpoints and 66 are not.

The $\mathsf{C}_{22}$ fraction 7/11 records 14 labelled slots out of 22. Slots 119–126 are unfilled in this reference; their assigned fixture values establish neither a physical cutoff nor an impossibility of further elements.

**Falsification.** The assignment table and its equations must agree exactly. Positive $X^{\mathrm{fix}}$ for negative $S^{\mathrm{fix}}$, or a changed assigned set without a new input binding, fails the fixture. Agreement with these definitions is not empirical validation.

*Provenance.* Main (107), Main (108), Main §5.7.12, Main §5.6.6.

### 4.3 Complete reference ledger

The authoritative generated ledger is `manual2/data/m1/matter1_elements_1_118_ledger.csv`. It binds each static declaration to its input record, distinguishes labelled elements from unfilled slots, and carries pending admission and unexecuted native status. The support declaration is syntax; the support sum and assigned fields retain the scopes above. Assigned route codes are `neutral`, `deficit`, `surplus` and `unfilled`; group-boundary and named-anchor flags are independent attributes. The generated table and figure consume the same validated data. Declared compatibility copies must compare equal.

Table 3: **Complete declared reference and conditional support ledger.** $Z = 1 \dots 118$ names supplied elements; slots $119 \dots 126$ are unfilled reference positions. Pressure and route classes are assigned inputs; the support sum is a conditional expression, not an admitted native duration. Grouping follows Main §5.7.13; the expression specializes Main (35).

| $Z$ | Sym | Ref. | $A$ | $\mathsf{C}_{2m}$ | Local $a{:}b{:}c$ | $u_-$ | $h$ | $u_+$ | $U$ | $P^{\mathrm{fix}}$ | $C^{\mathrm{fix}}$ | $S^{\mathrm{fix}}, X^{\mathrm{fix}}$ | Assigned class |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | H | 1 | | $\mathsf{C}_6$ | 1 : 0 : 1 | 30 | 1 | 4 | 35 | 15 | 18 | $-3, 0$ | Deficit |
| 2 | He | 4 | | $\mathsf{C}_6$ | 1 : 1 : 0 | 196 | 1 | 8 | 205 | 18 | 18 | $0, 0$ | Neutral |
| 3 | Li | 7 | | $\mathsf{C}_6$ | 2 : 1 : 1 | 362 | 1 | 12 | 375 | 18 | 18 | $0, 0$ | Neutral |
| 4 | Be | 9 | | $\mathsf{C}_6$ | 2 : 2 : 0 | 460 | 1 | 16 | 477 | 18 | 18 | $0, 0$ | Neutral |
| 5 | B | 11 | | $\mathsf{C}_6$ | 3 : 2 : 1 | 558 | 1 | 20 | 579 | 18 | 18 | $0, 0$ | Neutral |
| 6 | C | 12 | | $\mathsf{C}_6$ | 3 : 3 : 0 | 588 | 1 | 24 | 613 | 18 | 18 | $0, 0$ | Neutral + anchor |
| 7 | N | 14 | | $\mathsf{C}_8$ | 1 : 0 : 1 | 686 | 1 | 4 | 691 | 24 | 24 | $0, 0$ | Neutral |
| 8 | O | 16 | | $\mathsf{C}_8$ | 1 : 1 : 0 | 784 | 1 | 8 | 793 | 21 | 24 | $-3, 0$ | Deficit |
| 9 | F | 19 | | $\mathsf{C}_8$ | 2 : 1 : 1 | 950 | 1 | 12 | 963 | 21 | 24 | $-3, 0$ | Deficit |
| 10 | Ne | 20 | | $\mathsf{C}_8$ | 2 : 2 : 0 | 980 | 1 | 16 | 997 | 24 | 24 | $0, 0$ | Neutral |
| 11 | Na | 23 | | $\mathsf{C}_8$ | 3 : 2 : 1 | 1146 | 1 | 20 | 1167 | 24 | 24 | $0, 0$ | Neutral |
| 12 | Mg | 24 | | $\mathsf{C}_8$ | 3 : 3 : 0 | 1176 | 1 | 24 | 1201 | 24 | 24 | $0, 0$ | Neutral |
| 13 | Al | 27 | | $\mathsf{C}_8$ | 4 : 3 : 1 | 1342 | 1 | 28 | 1371 | 24 | 24 | $0, 0$ | Neutral |
| 14 | Si | 28 | | $\mathsf{C}_8$ | 4 : 4 : 0 | 1372 | 1 | 32 | 1405 | 24 | 24 | $0, 0$ | Neutral + anchor |
| 15 | P | 31 | | $\mathsf{C}_{10}$ | 1 : 0 : 1 | 1538 | 1 | 4 | 1543 | 30 | 30 | $0, 0$ | Neutral |
| 16 | S | 32 | | $\mathsf{C}_{10}$ | 1 : 1 : 0 | 1568 | 1 | 8 | 1577 | 27 | 30 | $-3, 0$ | Deficit |
| 17 | Cl | 35 | | $\mathsf{C}_{10}$ | 2 : 1 : 1 | 1734 | 1 | 12 | 1747 | 27 | 30 | $-3, 0$ | Deficit |
| 18 | Ar | 40 | | $\mathsf{C}_{10}$ | 2 : 2 : 0 | 2036 | 1 | 16 | 2053 | 30 | 30 | $0, 0$ | Neutral |
| 19 | K | 39 | | $\mathsf{C}_{10}$ | 3 : 2 : 1 | 1930 | 1 | 20 | 1951 | 30 | 30 | $0, 0$ | Neutral |
| 20 | Ca | 40 | | $\mathsf{C}_{10}$ | 3 : 3 : 0 | 1960 | 1 | 24 | 1985 | 30 | 30 | $0, 0$ | Neutral |
| 21 | Sc | 45 | | $\mathsf{C}_{10}$ | 4 : 3 : 1 | 2262 | 1 | 28 | 2291 | 30 | 30 | $0, 0$ | Neutral |
| 22 | Ti | 48 | | $\mathsf{C}_{10}$ | 4 : 4 : 0 | 2428 | 1 | 32 | 2461 | 30 | 30 | $0, 0$ | Neutral |
| 23 | V | 51 | | $\mathsf{C}_{10}$ | 5 : 4 : 1 | 2594 | 1 | 36 | 2631 | 30 | 30 | $0, 0$ | Neutral |
| 24 | Cr | 52 | | $\mathsf{C}_{10}$ | 5 : 5 : 0 | 2624 | 1 | 40 | 2665 | 30 | 30 | $0, 0$ | Neutral + anchor |
| 25 | Mn | 55 | | $\mathsf{C}_{12}$ | 1 : 0 : 1 | 2790 | 1 | 4 | 2795 | 36 | 36 | $0, 0$ | Neutral |
| 26 | Fe | 56 | | $\mathsf{C}_{12}$ | 1 : 1 : 0 | 2820 | 1 | 8 | 2829 | 36 | 36 | $0, 0$ | Neutral |
| 27 | Co | 59 | | $\mathsf{C}_{12}$ | 2 : 1 : 1 | 2986 | 1 | 12 | 2999 | 36 | 36 | $0, 0$ | Neutral |
| 28 | Ni | 58 | | $\mathsf{C}_{12}$ | 2 : 2 : 0 | 2880 | 1 | 16 | 2897 | 36 | 36 | $0, 0$ | Neutral |
| 29 | Cu | 63 | | $\mathsf{C}_{12}$ | 3 : 2 : 1 | 3182 | 1 | 20 | 3203 | 36 | 36 | $0, 0$ | Neutral |
| 30 | Zn | 64 | | $\mathsf{C}_{12}$ | 3 : 3 : 0 | 3212 | 1 | 24 | 3237 | 36 | 36 | $0, 0$ | Neutral |
| 31 | Ga | 69 | | $\mathsf{C}_{12}$ | 4 : 3 : 1 | 3514 | 1 | 28 | 3543 | 36 | 36 | $0, 0$ | Neutral |
| 32 | Ge | 74 | | $\mathsf{C}_{12}$ | 4 : 4 : 0 | 3816 | 1 | 32 | 3849 | 36 | 36 | $0, 0$ | Neutral |
| 33 | As | 75 | | $\mathsf{C}_{12}$ | 5 : 4 : 1 | 3846 | 1 | 36 | 3883 | 36 | 36 | $0, 0$ | Neutral |
| 34 | Se | 80 | | $\mathsf{C}_{12}$ | 5 : 5 : 0 | 4148 | 1 | 40 | 4189 | 33 | 36 | $-3, 0$ | Deficit |
| 35 | Br | 79 | | $\mathsf{C}_{12}$ | 6 : 5 : 1 | 4042 | 1 | 44 | 4087 | 33 | 36 | $-3, 0$ | Deficit |
| 36 | Kr | 84 | | $\mathsf{C}_{12}$ | 6 : 6 : 0 | 4344 | 1 | 48 | 4393 | 36 | 36 | $0, 0$ | Neutral + anchor |
| 37 | Rb | 85 | | $\mathsf{C}_{14}$ | 1 : 0 : 1 | 4374 | 1 | 4 | 4379 | 42 | 42 | $0, 0$ | Neutral |
| 38 | Sr | 88 | | $\mathsf{C}_{14}$ | 1 : 1 : 0 | 4540 | 1 | 8 | 4549 | 42 | 42 | $0, 0$ | Neutral |
| 39 | Y | 89 | | $\mathsf{C}_{14}$ | 2 : 1 : 1 | 4570 | 1 | 12 | 4583 | 42 | 42 | $0, 0$ | Neutral |
| 40 | Zr | 90 | | $\mathsf{C}_{14}$ | 2 : 2 : 0 | 4600 | 1 | 16 | 4617 | 42 | 42 | $0, 0$ | Neutral |
| 41 | Nb | 93 | | $\mathsf{C}_{14}$ | 3 : 2 : 1 | 4766 | 1 | 20 | 4787 | 42 | 42 | $0, 0$ | Neutral |
| 42 | Mo | 98 | | $\mathsf{C}_{14}$ | 3 : 3 : 0 | 5068 | 1 | 24 | 5093 | 42 | 42 | $0, 0$ | Neutral |
| 43 | Tc | 98 | | $\mathsf{C}_{14}$ | 4 : 3 : 1 | 5030 | 1 | 28 | 5059 | 45 | 42 | $3, 3$ | Surplus |
| 44 | Ru | 102 | | $\mathsf{C}_{14}$ | 4 : 4 : 0 | 5264 | 1 | 32 | 5297 | 42 | 42 | $0, 0$ | Neutral |
| 45 | Rh | 103 | | $\mathsf{C}_{14}$ | 5 : 4 : 1 | 5294 | 1 | 36 | 5331 | 42 | 42 | $0, 0$ | Neutral |
| 46 | Pd | 106 | | $\mathsf{C}_{14}$ | 5 : 5 : 0 | 5460 | 1 | 40 | 5501 | 42 | 42 | $0, 0$ | Neutral |
| 47 | Ag | 107 | | $\mathsf{C}_{14}$ | 6 : 5 : 1 | 5490 | 1 | 44 | 5535 | 42 | 42 | $0, 0$ | Neutral |
| 48 | Cd | 114 | | $\mathsf{C}_{14}$ | 6 : 6 : 0 | 5928 | 1 | 48 | 5977 | 42 | 42 | $0, 0$ | Neutral |
| 49 | In | 115 | | $\mathsf{C}_{14}$ | 7 : 6 : 1 | 5958 | 1 | 52 | 6011 | 42 | 42 | $0, 0$ | Neutral |
| 50 | Sn | 120 | | $\mathsf{C}_{14}$ | 7 : 7 : 0 | 6260 | 1 | 56 | 6317 | 42 | 42 | $0, 0$ | Neutral + anchor |
| 51 | Sb | 121 | | $\mathsf{C}_{16}$ | 1 : 0 : 1 | 6290 | 1 | 4 | 6295 | 48 | 48 | $0, 0$ | Neutral |
| 52 | Te | 130 | | $\mathsf{C}_{16}$ | 1 : 1 : 0 | 6864 | 1 | 8 | 6873 | 45 | 48 | $-3, 0$ | Deficit |
| 53 | I | 127 | | $\mathsf{C}_{16}$ | 2 : 1 : 1 | 6622 | 1 | 12 | 6635 | 45 | 48 | $-3, 0$ | Deficit |
| 54 | Xe | 132 | | $\mathsf{C}_{16}$ | 2 : 2 : 0 | 6924 | 1 | 16 | 6941 | 48 | 48 | $0, 0$ | Neutral |
| 55 | Cs | 133 | | $\mathsf{C}_{16}$ | 3 : 2 : 1 | 6954 | 1 | 20 | 6975 | 48 | 48 | $0, 0$ | Neutral |
| 56 | Ba | 138 | | $\mathsf{C}_{16}$ | 3 : 3 : 0 | 7256 | 1 | 24 | 7281 | 48 | 48 | $0, 0$ | Neutral |
| 57 | La | 139 | | $\mathsf{C}_{16}$ | 4 : 3 : 1 | 7286 | 1 | 28 | 7315 | 48 | 48 | $0, 0$ | Neutral |
| 58 | Ce | 140 | | $\mathsf{C}_{16}$ | 4 : 4 : 0 | 7316 | 1 | 32 | 7349 | 48 | 48 | $0, 0$ | Neutral |
| 59 | Pr | 141 | | $\mathsf{C}_{16}$ | 5 : 4 : 1 | 7346 | 1 | 36 | 7383 | 48 | 48 | $0, 0$ | Neutral |
| 60 | Nd | 142 | | $\mathsf{C}_{16}$ | 5 : 5 : 0 | 7376 | 1 | 40 | 7417 | 48 | 48 | $0, 0$ | Neutral |
| 61 | Pm | 145 | | $\mathsf{C}_{16}$ | 6 : 5 : 1 | 7542 | 1 | 44 | 7587 | 51 | 48 | $3, 3$ | Surplus |

| $Z$ | Sym | Ref. | $A$ | $\mathsf{C}_{2m}$ | Local $a{:}b{:}c$ | $u_-$ | $h$ | $u_+$ | $U$ | $P^{\mathrm{fix}}$ | $C^{\mathrm{fix}}$ | $S^{\mathrm{fix}}, X^{\mathrm{fix}}$ | Assigned class |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 62 | Sm | 152 | | $\mathsf{C}_{16}$ | 6 : 6 : 0 | 7980 | 1 | 48 | 8029 | 48 | 48 | 0, 0 | Neutral |
| 63 | Eu | 153 | | $\mathsf{C}_{16}$ | 7 : 6 : 1 | 8010 | 1 | 52 | 8063 | 48 | 48 | 0, 0 | Neutral |
| 64 | Gd | 158 | | $\mathsf{C}_{16}$ | 7 : 7 : 0 | 8312 | 1 | 56 | 8369 | 48 | 48 | 0, 0 | Neutral |
| 65 | Tb | 159 | | $\mathsf{C}_{16}$ | 8 : 7 : 1 | 8342 | 1 | 60 | 8403 | 48 | 48 | 0, 0 | Neutral |
| 66 | Dy | 164 | | $\mathsf{C}_{16}$ | 8 : 8 : 0 | 8644 | 1 | 64 | 8709 | 48 | 48 | 0, 0 | Neutral + anchor |
| 67 | Ho | 165 | | $\mathsf{C}_{18}$ | 1 : 0 : 1 | 8674 | 1 | 4 | 8679 | 54 | 54 | 0, 0 | Neutral |
| 68 | Er | 166 | | $\mathsf{C}_{18}$ | 1 : 1 : 0 | 8704 | 1 | 8 | 8713 | 54 | 54 | 0, 0 | Neutral |
| 69 | Tm | 169 | | $\mathsf{C}_{18}$ | 2 : 1 : 1 | 8870 | 1 | 12 | 8883 | 54 | 54 | 0, 0 | Neutral |
| 70 | Yb | 174 | | $\mathsf{C}_{18}$ | 2 : 2 : 0 | 9172 | 1 | 16 | 9189 | 54 | 54 | 0, 0 | Neutral |
| 71 | Lu | 175 | | $\mathsf{C}_{18}$ | 3 : 2 : 1 | 9202 | 1 | 20 | 9223 | 54 | 54 | 0, 0 | Neutral |
| 72 | Hf | 180 | | $\mathsf{C}_{18}$ | 3 : 3 : 0 | 9504 | 1 | 24 | 9529 | 54 | 54 | 0, 0 | Neutral |
| 73 | Ta | 181 | | $\mathsf{C}_{18}$ | 4 : 3 : 1 | 9534 | 1 | 28 | 9563 | 54 | 54 | 0, 0 | Neutral |
| 74 | W | 184 | | $\mathsf{C}_{18}$ | 4 : 4 : 0 | 9700 | 1 | 32 | 9733 | 54 | 54 | 0, 0 | Neutral |
| 75 | Re | 187 | | $\mathsf{C}_{18}$ | 5 : 4 : 1 | 9866 | 1 | 36 | 9903 | 54 | 54 | 0, 0 | Neutral |
| 76 | Os | 192 | | $\mathsf{C}_{18}$ | 5 : 5 : 0 | 10168 | 1 | 40 | 10209 | 54 | 54 | 0, 0 | Neutral |
| 77 | Ir | 193 | | $\mathsf{C}_{18}$ | 6 : 5 : 1 | 10198 | 1 | 44 | 10243 | 54 | 54 | 0, 0 | Neutral |
| 78 | Pt | 195 | | $\mathsf{C}_{18}$ | 6 : 6 : 0 | 10296 | 1 | 48 | 10345 | 54 | 54 | 0, 0 | Neutral |
| 79 | Au | 197 | | $\mathsf{C}_{18}$ | 7 : 6 : 1 | 10394 | 1 | 52 | 10447 | 54 | 54 | 0, 0 | Neutral |
| 80 | Hg | 202 | | $\mathsf{C}_{18}$ | 7 : 7 : 0 | 10696 | 1 | 56 | 10753 | 54 | 54 | 0, 0 | Neutral |
| 81 | Tl | 205 | | $\mathsf{C}_{18}$ | 8 : 7 : 1 | 10862 | 1 | 60 | 10923 | 54 | 54 | 0, 0 | Neutral |
| 82 | Pb | 208 | | $\mathsf{C}_{18}$ | 8 : 8 : 0 | 11028 | 1 | 64 | 11093 | 54 | 54 | 0, 0 | Neutral |
| 83 | Bi | 209 | | $\mathsf{C}_{18}$ | 9 : 8 : 1 | 11058 | 1 | 68 | 11127 | 54 | 54 | 0, 0 | Neutral |
| 84 | Po | 209 | | $\mathsf{C}_{18}$ | 9 : 9 : 0 | 11020 | 1 | 72 | 11093 | 57 | 54 | 3, 3 | Surplus + anchor |
| 85 | At | 210 | | $\mathsf{C}_{20}$ | 1 : 0 : 1 | 11050 | 1 | 4 | 11055 | 63 | 60 | 3, 3 | Surplus |
| 86 | Rn | 222 | | $\mathsf{C}_{20}$ | 1 : 1 : 0 | 11828 | 1 | 8 | 11837 | 63 | 60 | 3, 3 | Surplus |
| 87 | Fr | 223 | | $\mathsf{C}_{20}$ | 2 : 1 : 1 | 11858 | 1 | 12 | 11871 | 63 | 60 | 3, 3 | Surplus |
| 88 | Ra | 226 | | $\mathsf{C}_{20}$ | 2 : 2 : 0 | 12024 | 1 | 16 | 12041 | 63 | 60 | 3, 3 | Surplus |
| 89 | Ac | 227 | | $\mathsf{C}_{20}$ | 3 : 2 : 1 | 12054 | 1 | 20 | 12075 | 63 | 60 | 3, 3 | Surplus |
| 90 | Th | 232 | | $\mathsf{C}_{20}$ | 3 : 3 : 0 | 12356 | 1 | 24 | 12381 | 66 | 60 | 6, 6 | Surplus |
| 91 | Pa | 231 | | $\mathsf{C}_{20}$ | 4 : 3 : 1 | 12250 | 1 | 28 | 12279 | 66 | 60 | 6, 6 | Surplus |
| 92 | U | 238 | | $\mathsf{C}_{20}$ | 4 : 4 : 0 | 12688 | 1 | 32 | 12721 | 66 | 60 | 6, 6 | Surplus |
| 93 | Np | 237 | | $\mathsf{C}_{20}$ | 5 : 4 : 1 | 12582 | 1 | 36 | 12619 | 66 | 60 | 6, 6 | Surplus |
| 94 | Pu | 244 | | $\mathsf{C}_{20}$ | 5 : 5 : 0 | 13020 | 1 | 40 | 13061 | 66 | 60 | 6, 6 | Surplus |
| 95 | Am | 243 | | $\mathsf{C}_{20}$ | 6 : 5 : 1 | 12914 | 1 | 44 | 12959 | 66 | 60 | 6, 6 | Surplus |
| 96 | Cm | 247 | | $\mathsf{C}_{20}$ | 6 : 6 : 0 | 13148 | 1 | 48 | 13197 | 69 | 60 | 9, 9 | Surplus |
| 97 | Bk | 247 | | $\mathsf{C}_{20}$ | 7 : 6 : 1 | 13110 | 1 | 52 | 13163 | 69 | 60 | 9, 9 | Surplus |
| 98 | Cf | 251 | | $\mathsf{C}_{20}$ | 7 : 7 : 0 | 13344 | 1 | 56 | 13401 | 69 | 60 | 9, 9 | Surplus |
| 99 | Es | 252 | | $\mathsf{C}_{20}$ | 8 : 7 : 1 | 13374 | 1 | 60 | 13435 | 69 | 60 | 9, 9 | Surplus |
| 100 | Fm | 257 | | $\mathsf{C}_{20}$ | 8 : 8 : 0 | 13676 | 1 | 64 | 13741 | 69 | 60 | 9, 9 | Surplus |
| 101 | Md | 258 | | $\mathsf{C}_{20}$ | 9 : 8 : 1 | 13706 | 1 | 68 | 13775 | 69 | 60 | 9, 9 | Surplus |
| 102 | No | 259 | | $\mathsf{C}_{20}$ | 9 : 9 : 0 | 13736 | 1 | 72 | 13809 | 72 | 60 | 12, 12 | Surplus |
| 103 | Lr | 266 | | $\mathsf{C}_{20}$ | 10 : 9 : 1 | 14174 | 1 | 76 | 14251 | 72 | 60 | 12, 12 | Surplus |
| 104 | Rf | 267 | | $\mathsf{C}_{20}$ | 10 : 10 : 0 | 14204 | 1 | 80 | 14285 | 72 | 60 | 12, 12 | Surplus + anchor |
| 105 | Db | 268 | | $\mathsf{C}_{22}$ | 1 : 0 : 1 | 14234 | 1 | 4 | 14239 | 78 | 66 | 12, 12 | Surplus |
| 106 | Sg | 269 | | $\mathsf{C}_{22}$ | 1 : 1 : 0 | 14264 | 1 | 8 | 14273 | 78 | 66 | 12, 12 | Surplus |
| 107 | Bh | 270 | | $\mathsf{C}_{22}$ | 2 : 1 : 1 | 14294 | 1 | 12 | 14307 | 78 | 66 | 12, 12 | Surplus |
| 108 | Hs | 277 | | $\mathsf{C}_{22}$ | 2 : 2 : 0 | 14732 | 1 | 16 | 14749 | 81 | 66 | 15, 15 | Surplus |
| 109 | Mt | 278 | | $\mathsf{C}_{22}$ | 3 : 2 : 1 | 14762 | 1 | 20 | 14783 | 81 | 66 | 15, 15 | Surplus |
| 110 | Ds | 281 | | $\mathsf{C}_{22}$ | 3 : 3 : 0 | 14928 | 1 | 24 | 14953 | 81 | 66 | 15, 15 | Surplus |
| 111 | Rg | 282 | | $\mathsf{C}_{22}$ | 4 : 3 : 1 | 14958 | 1 | 28 | 14987 | 81 | 66 | 15, 15 | Surplus |
| 112 | Cn | 285 | | $\mathsf{C}_{22}$ | 4 : 4 : 0 | 15124 | 1 | 32 | 15157 | 81 | 66 | 15, 15 | Surplus |
| 113 | Nh | 286 | | $\mathsf{C}_{22}$ | 5 : 4 : 1 | 15154 | 1 | 36 | 15191 | 81 | 66 | 15, 15 | Surplus |
| 114 | Fl | 289 | | $\mathsf{C}_{22}$ | 5 : 5 : 0 | 15320 | 1 | 40 | 15361 | 84 | 66 | 18, 18 | Surplus |
| 115 | Mc | 290 | | $\mathsf{C}_{22}$ | 6 : 5 : 1 | 15350 | 1 | 44 | 15395 | 84 | 66 | 18, 18 | Surplus |
| 116 | Lv | 293 | | $\mathsf{C}_{22}$ | 6 : 6 : 0 | 15516 | 1 | 48 | 15565 | 84 | 66 | 18, 18 | Surplus |
| 117 | Ts | 294 | | $\mathsf{C}_{22}$ | 7 : 6 : 1 | 15546 | 1 | 52 | 15599 | 84 | 66 | 18, 18 | Surplus |
| 118 | Og | 294 | | $\mathsf{C}_{22}$ | 7 : 7 : 0 | 15508 | 1 | 56 | 15565 | 84 | 66 | 18, 18 | Surplus |
| 119 | – | 297 | | $\mathsf{C}_{22}$ | 8 : 7 : 1 | 15674 | 1 | 60 | 15735 | 69 | 66 | 3, 3 | Unfilled fixture |
| 120 | – | 298 | | $\mathsf{C}_{22}$ | 8 : 8 : 0 | 15704 | 1 | 64 | 15769 | 72 | 66 | 6, 6 | Unfilled fixture |
| 121 | – | 301 | | $\mathsf{C}_{22}$ | 9 : 8 : 1 | 15870 | 1 | 68 | 15939 | 75 | 66 | 9, 9 | Unfilled fixture |
| 122 | – | 302 | | $\mathsf{C}_{22}$ | 9 : 9 : 0 | 15900 | 1 | 72 | 15973 | 78 | 66 | 12, 12 | Unfilled fixture |
| 123 | – | 305 | | $\mathsf{C}_{22}$ | 10 : 9 : 1 | 16066 | 1 | 76 | 16143 | 81 | 66 | 15, 15 | Unfilled fixture |
| 124 | – | 306 | | $\mathsf{C}_{22}$ | 10 : 10 : 0 | 16096 | 1 | 80 | 16177 | 84 | 66 | 18, 18 | Unfilled fixture |
| 125 | – | 309 | | $\mathsf{C}_{22}$ | 11 : 10 : 1 | 16262 | 1 | 84 | 16347 | 87 | 66 | 21, 21 | Unfilled fixture |
| 126 | – | 310 | | $\mathsf{C}_{22}$ | 11 : 11 : 0 | 16292 | 1 | 88 | 16381 | 90 | 66 | 24, 24 | Unfilled fixture |

# 5 Verification Scope and Pending Native Evidence

## 5.1 Executed static checks and inherited audit obligations

**Setup.** The static checker verifies stored inputs and generated artifacts without rewriting them. Generation is a separate explicit operation. Exact grouping, local tuples, support sums, assignment residuals, identities and data/figure agreement are testable here. Their success does not certify the six native hooks in Main §5.17.

**Derivation.** Table 4 separates the available static checks from the instance evidence required for native claims. A check result is reported only after that check executes; this table describes scope, not an unrecorded run count.

Table 4: **Verification scope.** Element-wide native obligations remain pending.

| Inherited hook | Static reference check | Native evidence not supplied |
|---|---|---|
| Spine | Declared role mapping and operand sum | Retained occurrences at each finite depth |
| Seat/enclosure | Nine boundaries, tuples and labelled occupancy | Scoped field-family and routing witness |
| Sheddic path | Assigned class, route and residual consistency | Realized path and retention window |
| Cycle | Support syntax and pending status | Construction, retained start and return certificate |
| Fusion | Conditional operand and residual equations | Admitted inner forms and outer closure certificate |
| Naming | Stable declarations, record kinds and input bindings | No name substitutes for instance evidence |

**Specialization.** A future native D.E.C. record has the inherited form

$$(B,\ v,\ e,\ v_e,\ \sigma(e),\ \mathrm{adm}(e;B),\ w(e;B),\ P(e;B),\ \mathrm{route}(e;B)).$$

The present element reference ledger does not instantiate it. The separately declared finite model in §6 executes its own documented predicates and histories; it is not an element-wide D.E.C. ledger. Native execution would require bound $Q_4$ Hamming-1 traversals, scope and window, admissibility and exact weights, normalized kernels on nonempty positive-weight support, and reversed-edge SADAR pairs:

$$\sum_e P(e;B) = 1, \qquad \mathrm{adm}(e;B) = 0 \Rightarrow P(e;B) = 0, \qquad \mathfrak{S}_B(C_{\bar{e}}) = -\mathfrak{S}_B(C_e).$$

An empty admissible support does not define a normalized kernel. Each retained-return witness must bind its own retained identity, scope and lineage. Fold and comparison consumers must additionally preserve explicit compatible maps between their distinct scopes and lineages. None of these obligations is discharged by this reference's assignment tables.

**Falsification.** Actual negative probes must reject changed values, duplicate or missing identities, mismatched mirrors, invalid status promotion and altered provenance without replacing the input bytes. In particular, a forged certificate-like string must not turn pending admission into a pass. Unknown command flags must fail, and the verification command must not regenerate corrupted input.

Publication checks bind Main source/PDF/AUX/FLS before resolving semantic references. Figure checks cover all 126 marks at final size, including the eight unfilled slots. These are distinct from native dynamics or empirical tests.

*Provenance.* Main §5.17.1, Main §5.17.4, Main §5.17.5, Main §5.17.7, Main §5.17.8, Main §5.17.10, Main (66), Main (95).

*Literature note.* D.E.C. row grammar and execution protocol: released AOD Manual v41.02, §4 [8].

# 6 Folded Construction and Relational Time

## 6.1 Distinction, handed hinge and retained lineage

**Setup.** The schematic sameness presentation $\cdots xxx \cdots = x$ belongs to Main §1.1. Its occurrences are not previously distinguished events or ticks. The first distinction and its anchor precede the finite construction below. On an oriented construction, the shorthand $x(x)x$ resolves as $x^{-}(x^{0})x^{+}$: branch, retained hinge and counterbranch. Reflection exchanges the sides and preserves the hinge. This Manual instantiates those relations; it does not replace Main's ontology with a runtime string-reduction rule.

**Derivation.** Handedness records branch order through the hinge. The support accessor inherited from Main (31) is

$$RD_B(C) = \Pi_C^- + H_{\mu,C} + \Pi_C^+.$$

Equal scalar support does not identify oppositely oriented constructions. A duration realization, a count of traversals and a count of retained returns require their respective maps. In particular, a structural hinge is mandatory even when optional hinge dwell is zero; it need not be a seventh Hamming-1 traversal.

Folding relates distinct inner and outer scopes through explicit maps

$$f_- : (B_-, i_-, \Gamma_-) \longrightarrow (B_o, i_o), \qquad f_+ : (B_+, i_+, \Gamma_+) \longrightarrow (B_o, i_o).$$

These maps preserve the source identity, scope, lineage, orientation, history and certificate. A proposed outer boundary requires its own retained start and return witness $\Gamma_o$. Exposure follows the maps back to the inner records. Neither common endpoint coordinates nor a handedness flip supplies an outer certificate. Cross-scale scope equality is not required.

**Provenance.** The source obligations are Main (5), Main (6), Main (46), Main (41) and Main (42).

## 6.2 An injective two-face example

**Setup.** Declare six distinct vertices of $Q_4$, with each committed base traversal changing one coordinate:

| vertex | $A$ | $B$ | $C$ | $D$ | $E$ | $F$ |
|---|---|---|---|---|---|---|
| address | 0000 | 1000 | 1100 | 0100 | 0010 | 1010 |

Take the oriented square faces $C_- = (A, B, C, D, A)$ and $C_+ = (B, A, E, F, B)$. They share only the coordinate edge $AB$, with opposite orientation. Assign interface roles to $A, B$, negative-side roles to $C, D$, and positive-side roles to $E, F$. The role projection is separate from the coordinate path.

**Derivation.** Cancellation in the oriented boundary chain gives

$$\partial(C_- + C_+) = (A, E, F, B, C, D, A).$$

The shared interface remains present in the incidence and history records. Cancellation of its outer coefficient does not erase that record. The two legs $(A, E, F, B)$ and $(B, C, D, A)$ each contain three Hamming-1 traversals. For $n$ declared repetitions, let $x$ denote the number of committed traversals in one leg family. Then

$$x = 3n, \qquad 2x = 6n. \tag{18}$$

Here $n$ counts repetitions of this specified pattern; one counted tick is one committed Hamming-1 traversal. The factor three follows from the enumerated edges, not from naming three roles. The persisted illustrative ledger contains three repetitions, nine traversals per leg and eighteen distinct event identities. It asserts no native closure or beat count.

**Specialization.** Main's two-nonempty-leg bound in Main (28) constrains an already admitted closure. It neither selects this six-edge construction nor fixes a universal duration. A four-edge square is also a closed Hamming-1 walk. Conversely, closure of a geometric walk alone does not admit a native motif. One accepted retained return contributes one closure phase; later distinct returns of that same lineage supply beats as specified by Main (21). Replaying a record supplies neither a new traversal nor a new return.

If an additional model identifies each branch support operand with $3n$, its RD expression is $6n + H_\mu$, with $6n+1$ only in the single-hinge specialization. That support statement is distinct from the six-edge traversal count. The mathematical constant $\pi$, physical seconds and an atomic frequency do not enter this finite counting model.

**Falsification.** The checker rejects the coordinate aliases $E = D$, $F = C$ before counting: they make the face boundaries coincide with opposite orientation rather than leave the claimed six-edge boundary. It also rejects a reversed interface, a non-Hamming-1 edge, a missing role, duplicate external event identities, reserialized/recounted events, wrong scope or lineage, and unsupported status promotion. Checks run against persisted inputs without changing their bytes. The generator is a separate command.

*Provenance.* `manual2/data/m1/folding/two_square_fold.json`, `event_ledger.json` and `folding_report.json`; Main (8), Main (73) and Main (15).

### 6.3 Declared recurrence and mapped comparison

**Setup.** The separate finite model commits the next oriented edge at each update cut. A distinct start sample follows a completed return; optional hinge dwell contributes a zero-edge sample. These update cuts are the declared model's ordering domain, not seconds or a newly derived universal clock. Two inner square lineages, each with four retained returns, are embedded into each of two oppositely oriented outer lineages. The outer traces follow the six-edge boundary with their own starts and returns. Their inner maps remain explicit.

The model declares zero optional dwell for the positive outer orientation and one sample for the negative orientation. It declares phase biases $b_+ = 1/4$, $b_- = -1/4$, four comparison contacts, and three reporting cuts. These are fixture inputs. No target frequency, desired physical period or stochastic kernel is read by the generator.

**Derivation.** The inner completion cuts are $4, 9, 14, 19$. The positive outer completion cuts are $26, 33, 40, 47, 54$; the negative outer cuts are $27, 35, 43, 51, 59$. Thus both outer traces have six Hamming-1 traversals and the same declared support sum $6 + 1 = 7$, while their completion intervals are respectively seven and eight update cuts. Support and recurrence intervals retain different types even where their numbers coincide.

Each return witness binds its own start, hinge, events, scope and lineage. Event and return identities are unique; the first retained return supplies the origin, and subsequent returns supply beats. Two identity-preserving maps bind committed return prefixes to the comparator scope. Repeated contacts are required. For the windows in Table 5, define

$$\mathrm{biz}_{+|-} = \frac{\Delta \,\mathrm{bip}_+}{\Delta \,\mathrm{bip}_-}, \qquad \psi = b_+ - b_- = \frac{1}{2}, \qquad \Lambda(v) = N_+(v) - N_-(v) + \psi.$$

Here $N_\pm(v)$ are cumulative reclosure counts from each lineage's origin; they equal the window beat increments for these windows starting before both outer origins. A later window does not reset unwrapped lag. The reference counts are positive. Exchanging the comparator direction gives reciprocal cadence and negated bias/lag. Equal cadence in the first window does not remove its nonzero bias or lag.

Table 5: **Executed declared-model comparisons.** Drift is the change from the preceding reported unwrapped lag; its first value is undefined.

| Window | Beat counts $(+,-)$ | Cadence ratio | Unwrapped lag | Drift |
|---|---|---|---|---|
| $(20, 43]$ | $(2, 2)$ | $1$ | $1/2$ | undefined |
| $(20, 47]$ | $(3, 2)$ | $3/2$ | $3/2$ | $1$ |
| $(20, 54]$ | $(4, 3)$ | $4/3$ | $3/2$ | $0$ |

**Specialization.** The declared contact support is two traversals plus one structural hinge, giving three support units. An explicit additive contact-duration map adds reflection counts $(1, 1)$ in one direction and $(2, 2)$ in the other. Its directional duration counts are five and seven. These are contact durations under a declared map, not the outer support accessor or empirically measured periods. The model executes finite Hamming-1, hinge, own-return and fold-map predicates, while complete motif admission and native atomic execution remain pending.

*Provenance.* `temporal_model.json` declares the update and comparison inputs; `temporal_trace.json` stores events, witnesses, maps and contacts. Source anchors: Main (54), Main (55), Main (58), Main (61) and Main (63).

### 6.4 Causal returned-current response and exposure

**Setup.** For the same committed comparison histories, declare boundary weight $C_3 = 1$, oriented current $i = 1$ and forward duration participation $\rho^D_\omega = 5$. Use the preceding signed lag in the inherited returned-current expression. The three compatible capacities are declared inputs $10, 10, 3$, not predicted material properties. Responses commit strictly after the reporting cut that supplied their inputs.

**Derivation.** This specialization evaluates

$$\mathfrak{S} = C_3 \rho^D_\omega \Lambda i, \qquad P^{\mathfrak{S}} = C_3 \rho^D_\omega |\Lambda|, \qquad X = \max(0, P^{\mathfrak{S}} - C^{\mathrm{close}}).$$

It yields $\mathfrak{S} = 5/2, 15/2, 15/2$, residuals $-15/2, -5/2, 9/2$, and excesses $0, 0, 9/2$. The first two windows have nonzero lag and no positive excess. A separately bound reversal pair preserves boundary, window, history and duration participation while reversing edge orientation and signed current; its returned-current values are antisymmetric. This pairing is distinct from exchanging the two directional duration maps.

The responses commit at cuts $44, 48, 55$. The last loses positive-outer retention under the declared capacity rule, records an excess route and exposes the preserved inner maps and certificates. This is a model bridge between a typed excess and loss of outer retention; it is not an identity valid for every boundary. Routed-material reclosure and any radiation identification remain pending.

**Falsification.** Persisted tests reject forged finite certificates, missing structural hinge, omitted/substituted inner maps, recycled inner returns, replayed or repackaged returns, one-contact comparisons, forced duration symmetry, same-step self-causation, mismatched reversed-edge witnesses and unsupported status changes. The finite model's exact reproduction is tested independently of native admission. A source-derived dynamics rule, complete motif/accessor admission and an atomic observation map are still required before these fixture cadences or capacities can become physical predictions.

*Provenance.* `temporal_report.json` separates declared execution, finite witness scope, native admission and empirical status. Source anchors: Main (93), Main (95), Main (100), Main (107) and Main (108).

*Literature note.* The inherited distinction, fold and recurrence definitions are in the released AOD foundation [8]. The finite fixture and update model in this section are Manual applications.

# 7 Staged Hyperfine Application

## 7.1 Atomic identity and sourced recurrence

**Setup.** The proposed application compares recurrent atomic and oscillator systems through one common fusion/reclosure model. Element placement alone does not identify a hyperfine state: hold $Z$, isotope $A$, charge, electronic state and nuclear spin fixed, and distinguish the coupled states. The present inventory has two neutral isotopes, four states and two transitions. It adds no placement to the 118-element reference and changes no reference-A input.

Table 6: **External atomic inventory.** Frequencies are source records, not predictions of the finite folding model. Both clock transitions have $m_F = 0$ at each endpoint.

| Nuclide | Electronic state; $I, J$ | Transition | Frequency record |
|---|---|---|---|
| $^{1}$H, $Z = 1$ | $1s\,^2S_{1/2}$; $1/2, 1/2$ | $F = 1 \leftrightarrow 0$ | 1 420 405 751.768 Hz; reported one-sigma uncertainty 0.002 Hz; historical 1970 measurement |
| $^{133}$Cs, $Z = 55$ | [Xe]$6s\,^2S_{1/2}$; $7/2, 1/2$ | $F = 4 \leftrightarrow 3$ | 9 192 631 770 Hz; exact defining value; realization uncertainty is a separate quantity |

**Derivation.** Hellwig et al.'s printed page 200 reports the hydrogen value with uncertainty $2 \times 10^{-3}$ Hz. The exponent is bound to the retained page image, correcting the earlier plan's 0.02 Hz transcription. This is a dated measurement, not a current recommended value. Dividing the quoted decimal by the

defining caesium value gives the external reference ratio

$$r^{\mathrm{ref}}_{\mathrm{H/Cs}} = 0.1545156803086000, \qquad u_r^{\mathrm{reported}} \simeq 2.1757 \times 10^{-13}.$$

The implementation retains exact rational arithmetic of the transcribed decimals; the measured hydrogen frequency is not physically exact. The uncertainty shown divides the paper's reported one-sigma uncertainty by the exact defining denominator. It is not a new uncertainty evaluation or an independent recombination of the two hydrogen evaluations.

The inventory retains the common NBS-III caesium reference and the paper's wall, cavity-pulling, spin-exchange, magnetic-field and second-order Doppler correction context. Quantitative covariance and separate realization uncertainties remain unresolved where not supplied. A missing quantity is recorded as unknown, not zero. A new precision comparison must supply the applicable conditions and shared-reference uncertainty budget.

**Specialization.** Atomic interrogation prepares coherence and compares its later phase with an oscillator; population measurements provide feedback to that oscillator. The counted recurrence is the stabilized oscillator cycle. A detection count, Rabi oscillation, spontaneous waiting time and hyperfine frequency are different observables. Relating oscillator cycles and quantum phase to certified native reclosures requires a physical observation map.

*Provenance.* `manual2/data/m1/hyperfine/target_inventory.json` binds nuclide, state, transition, source and uncertainty records. `sources/hellwig1970-page200.png` preserves the transcription evidence. The report retains the ratio's measured-input scope.

*Literature note.* Historical measurement and state labels: Hellwig et al. [2]. Atomic data: NIST [4, 9]. Defining reference: BIPM [1]. Clock operation: NIST [3].

### 7.2 A common model and its remaining observation map

**Setup.** The application hypothesis is a reconfiguration between two states of the same isotope, bound to one frozen fusion/reclosure model. Proposed before/after configurations must preserve the isotope identity and the actual retained inner lineages. A transition need not increase element number, mass number or folding depth. A detected transfer can be identified as native shedding only after its source, destination, direction, window and measured observable are mapped.

**Derivation.** The atomic inventory (P4a), applicable construction (P2) and coupled recurrence (P3) are all prerequisites of the physical observation map (P4):

$$(\mathrm{P4a}, \mathrm{P2}, \mathrm{P3}) \longrightarrow \mathrm{P4}.$$

The finite declared-model checks of §6 instantiate selected construction and comparator predicates. They do not supply isotope-specific native admission, a physical state/configuration map, or an independently derived update rule. Thus P4 remains pending. Chirality or a ternary role cannot be equated with $m_F$ by visual resemblance, and a separate fitted law for each spectral line cannot count as a prediction of one common model.

The next physical comparison requires admitted atomic configurations, their recurrent histories and coupling records, a map from measured phase/cycles to native phase/returns, a metrological duration or frequency map, and its calibration and uncertainty model. Until those premises are established,

predicted frequency ratio and empirical score remain absent. The same restriction applies to a proposed occurrence of the mathematical constant $\pi$: no physical result is inferred from its use in a drawing or angular convention.

**Specialization.** The hydrogen target was exposed during the discussion and supplied plans before this implementation. `target_exposure.json` records exposure by 26 October 2026; an earlier exact first-exposure time is unknown. The known pilot is retrospective. A subsequent model freeze does not establish independent selection or blindness. A calibrant cannot also be scored as a held-out target. Independent prediction requires a separately justified target and selection history. Frequency-only data do not test phase bias, lag or drift.

**Falsification.** Persisted hostile tests reject changed atomic foreign keys, the incorrect hydrogen uncertainty, omitted correction lists and covariance statuses, unsupported native or empirical promotion, a policy allowing target access during candidate selection, a calibrant labelled held out, detection/cycle substitution, and per-line fitted models advertised as independent predictions. These are actual schema and provenance guard tests. Their success is not empirical agreement. Every consumed status is checked; missing statuses fail rather than silently becoming zero or success.

*Provenance.* `observation_map.json` records the P4a/P2/P3 dependencies and pending physical map. `hyperfine_report.json` lists unresolved dependencies and null prediction/score fields. Native recurrence, comparator and duration definitions are inherited from Main (61), Main (63) and Main (57).

*Literature note.* External atomic observations remain downstream of the released AOD definitions [8]; the historical target remains bound to Hellwig et al. [2].

## 8 Reference, History and Pending-Comparison Registry

**Setup.** This registry distinguishes active static calculations, a declared finite model, external source records, preserved historical declarations and pending native or measured comparisons. It does not classify every integer table as a completed native simulation.

Table 7: **Matter I evidence inventory.**

| Record | Present scope | Completion boundary |
|---|---|---|
| Support polynomial | Exact integer expression | Does not admit a motif |
| Enclosure groups and local tuples | Exact grouping and role projection | Does not execute a path |
| Reference-A and assigned classes | Pinned candidate fixture inputs | Original selection provenance unresolved |
| Conditional support and residuals | Exact consequences of those inputs | Native accessor/retention claims pending |
| v94 placement crosswalk | 118 rows, six compared fields | 708 comparisons; separate from lane inventory |
| v94 observable inventory | 118 elements across E0–E5 | Preserve original per-lane statuses |
| r24 occurrence/packet crosswalk | Historical declaration identities | Native transition and SADAR execution deferred |
| Element-wide D.E.C. and certificates | Required schema and obligations | Not executed; admission pending |
| Two-face construction | Injective Q4 incidence; external event ledger | Illustrative traversal count only |
| Finite fold and recurrence model | Mapped histories, own returns, coupled comparisons, causal response | Declared model executed; native admission pending |
| Hydrogen/caesium inventory | Two isotopes, four states, two transitions; historical ratio | Source inventory verified; physical map and scores pending |
| Other isotope, mass and spectral lanes | Future sourced inputs and comparisons | No element-wide catalog or scores supplied |

**Specialization.** The original v94 observable matrix has 708 rows: 118 E0 identity/registry rows marked `active_registry`, and 118 rows in each of E1 placement/classification, E2 atomic mass/standard atomic weight, E3 isotope support, E4 spectra and E5 reaction/routes marked `declared_lane_not_scored_in_v94`. Its ten value maps remain `deferred_not_attached`. These statuses are separate from the successful six-field placement comparison.

The r24 crosswalk retains occurrence, transition-packet and SADAR declaration IDs as historical IDs. Its execution remains declared but unexecuted; target values are not read, measured reports are closed and scores absent. It does not contain source-to-destination event witnesses. Correspondence of the original and public atlas is bound by parsed manifest content and five matching file hashes; it is not an identity claim about an entire public repository.

**Falsification.** A historical pending lane promoted to a current execution result without new evidence, or a hash comparison described as scientific admission, violates the inventory. Preserve predecessor status and current status in distinct fields. Prerecorded counterfactual outcomes do not count as actual hostile tests.

**Provenance.** The release provenance records the reviewed PR, original v94/r24 archive hashes, extracted member hashes, the six-field mapping, historical identifier joins and the current input bindings. Small verified source extracts and the generated `declaration_crosswalk.csv`, `observable_coverage.csv` and `value_map_status.csv` are stored under `manual2/data/m1/history/`. Their bindings are in `manual2/evidence/historical_sources.json`; current input hashes are in `manual2/evidence/`

`input_manifest.json`. Exact static checks, source identity checks and any future scientific claims retain separate result scopes.

# A Exact Arithmetic Fixtures

## A.1 Scope

These are arithmetic fixtures. They do not claim simulated native trajectories or certified closure. The inherited four-step convention is in Main App. I; polynomial support is in Main App. H.

## A.2 Four-step decompositions

For $L = 4c_L + r_L$, $r_L \in \{0, 1, 2, 3\}$, the outer even indices give

$$\begin{aligned} 6 &= 4(1) + 2, & 8 &= 4(2) + 0, & 10 &= 4(2) + 2, \\ 12 &= 4(3) + 0, & 14 &= 4(3) + 2, & 16 &= 4(4) + 0, \\ 18 &= 4(4) + 2, & 20 &= 4(5) + 0, & 22 &= 4(5) + 2. \end{aligned}$$

Even $L$ has remainder 0 or 2; an odd index has remainder 1 or 3. This division identity is not a theorem that every closure has four steps. The inherited history-weighted load requires its declared histories and weights (Main (172)).

## A.3 Labelled-reference occupancy

The 126 declared slots and 118 element labels give

$$\frac{118}{126} = \frac{59}{63}, \qquad \frac{N_{\mathrm{labelled}}(\mathsf{C}_{22})}{22} = \frac{14}{22} = \frac{7}{11}, \qquad \frac{N_{\mathrm{unfilled}}(\mathsf{C}_{22})}{22} = \frac{8}{22} = \frac{4}{11}.$$

These are exact occupancy fractions, independent of assigned TTL values and native retention.

## A.4 Falsification

Non-integer quotient/remainder, an invalid remainder, or disagreement with the labelled-slot counts fails the corresponding arithmetic fixture. It does not by itself falsify a native execution that has not been performed.

*Provenance.* Main App. I.2, Main App. I.4, (10).

# B Pending External Comparison

## B.1 Scope and numeric policy

No native-versus-measured score is executed in this release. The external inventory in §7 records a historical hydrogen measurement and the caesium defining reference; reference arithmetic is separate

from a model prediction. A future comparison must bind an established candidate, its declared value map and a sourced target record before evaluating

$$\mathsf{Q}_{\mathrm{ext}}(\nu_{Z,A\leftrightarrow\Omega};\mathcal{T}_Z). \tag{19}$$

A conditional support sum does not become a measured prediction merely by being frozen. Candidate selection and the mathematical/physical meaning of the value map must be explicit before joining targets.

External mass, spectral or isotope data retain their sources and units. A displayed measured decimal requires

$$(x_{\mathrm{disp}}, u_x, \mathrm{unit}(x), \mathrm{src}(x), \mathrm{round}(x), \mathrm{scope}(x)). \tag{20}$$

These records are distinct from the exact integer reference A and from $(Z, A-Z)$. AME and NIST references below identify possible comparison sources; no per-element selection of the present A inputs is attributed to them.

### B.2 Pending comparison gates

A future measured lane must identify the candidate and frozen inputs, applicable admission/execution evidence, value map, target version and row identity, units and precision, comparison formula, and resulting status. Missing inputs remain explicitly pending. No spectral, isotope-stability, charge, decay or reaction agreement is inferred from the reference figure.

*Provenance.* Main §5.16.12; released AOD Manual v41.02, §1.7–§1.8 and App. B.

*Literature note.* Potential external comparison sources: Wang et al. [10]; NIST ASD [5].

## C Exact Parity, Role and Occupancy Comparators

### C.1 Integer comparators

On each complete reference group $\mathsf{C}_{2m}$, $m = 3, \ldots, 11$:

1. Exactly $m$ seats have odd local parity and $m$ even local parity. Against the comparator $\widehat{O}_m = m$,

$$\Delta^{\mathbb{Z}}_{\mathrm{parity},m} = \widehat{O}_m - O_{\mathrm{even},m} = 0.$$

   Parity does not determine the assigned operational class.

2. The declared role projection has $m-1$ first-branch seats, one hinge seat, $m-1$ counterbranch seats and one boundary seat. Thus

$$\Delta^{\mathbb{Z}}_{\mathrm{branch},m} = (m-1) - (m-1) = 0.$$

   These are counts of visual roles, not traversed edge events.

3. The terminal group has 14 labelled slots and capacity 22:

$$\Delta^{\mathbb{Z}}_{\mathsf{C}_{22}} = 14 - 22 = -8.$$

   The difference counts eight unfilled reference slots and is not a retention or physical-cutoff residual.

### C.2 Signed-magnitude record

The comparator representation is

$$\Delta_k^{\mathbb{Z}} = s_k^{(3)} m_k, \qquad s_k^{(3)} \in \{-1, 0, +1\}, \quad m_k = |\Delta_k^{\mathbb{Z}}|.$$

The two balanced comparisons give $\delta_3 = (0, 0)$; the terminal occupancy comparison gives $\delta_3 = (-1, 8)$. These values express the declared integer counts, not a native phase-cycle measurement.

*Provenance.* (11), (12), Main §5.7.13; released AOD Manual v41.02, App. C, for the signed comparator convention.

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[10] M. Wang, W. J. Huang, F. G. Kondev, G. Audi, and S. Naimi. The AME 2020 atomic mass evaluation (II). *Chinese Physics C*, 45:030003, 2021.